

ELECTRICAL SCADA SYSTEM

Advanced Supervisory Control and Data Acquisition

for Smart Power Infrastructure

PROCESS STEPS

1 Data Collection



- Sensors, meters, relays, RTUs gather electrical parameters
- Voltage, current, frequency, power factor

2 Field Device Monitoring



- Circuit breakers
- Transformers
- Switchgear
- Capacitor banks

3 Communication Network



- Fiber optics
- Ethernet
- Wireless communication
- Industrial protocols (IEC 61850, DNP3, Modbus)

4 Remote Terminal Units (RTUs)



- Collect and transmit field data
- Real-time status updates

5 PLC Integration



- Programmable logic controllers
- Automated control operations

6 Central SCADA Server



- Data processing and storage
- Alarm management

7 Real-Time Visualization



- HMI dashboards
- Graphical monitoring screens

8 Automated Control Actions



- Remote switching
- Load balancing
- Fault isolation

9 Predictive Maintenance



- Asset health monitoring
- Failure prediction analytics

10 Grid Optimization



- Energy efficiency improvement
- Demand response management

SMART ELECTRICAL SUBSTATION



REAL-TIME MONITORING

- Sensors transmitting data
- Wireless signal icons
- Digital communication paths

COMMUNICATION NETWORK



RTUs & PLCs



SCADA CONTROL CENTER



AUTOMATED GRID CONTROL



CLOUD ANALYTICS PLATFORM

- AI-powered analytics dashboard
- Historical data storage
- Performance reports
- Predictive insights

POWER DISTRIBUTION NETWORK



RENEWABLE ENERGY INTEGRATION



BENEFITS OF ELECTRICAL SCADA

OPERATIONAL BENEFITS

- Real-Time Monitoring
- Faster Fault Detection
- Remote Operations
- Improved Reliability
- Reduced Downtime

PERFORMANCE BENEFITS

- Enhanced Grid Stability
- Better Asset Utilization
- Energy Optimization
- Reduced Power Losses
- Increased Efficiency

BUSINESS BENEFITS

- Lower Maintenance Costs
- Improved Safety
- Regulatory Compliance
- Predictive Maintenance
- Higher ROI

RELIABLE POWER MANAGEMENT

REAL-TIME GRID INTELLIGENCE

EFFICIENT & AUTOMATED OPERATIONS

APPLICATIONS



Electrical Substations



Smart Grids



Renewable Energy Plants



Industrial Manufacturing



Utility Power Networks



Water Treatment Plants



Oil & Gas Facilities



Data Centers



Railway Electrification



Building Energy Management

DATA FLOW

CONTROL FLOW

COMMAND / ACTION FLOW

WIRELESS COMMUNICATION